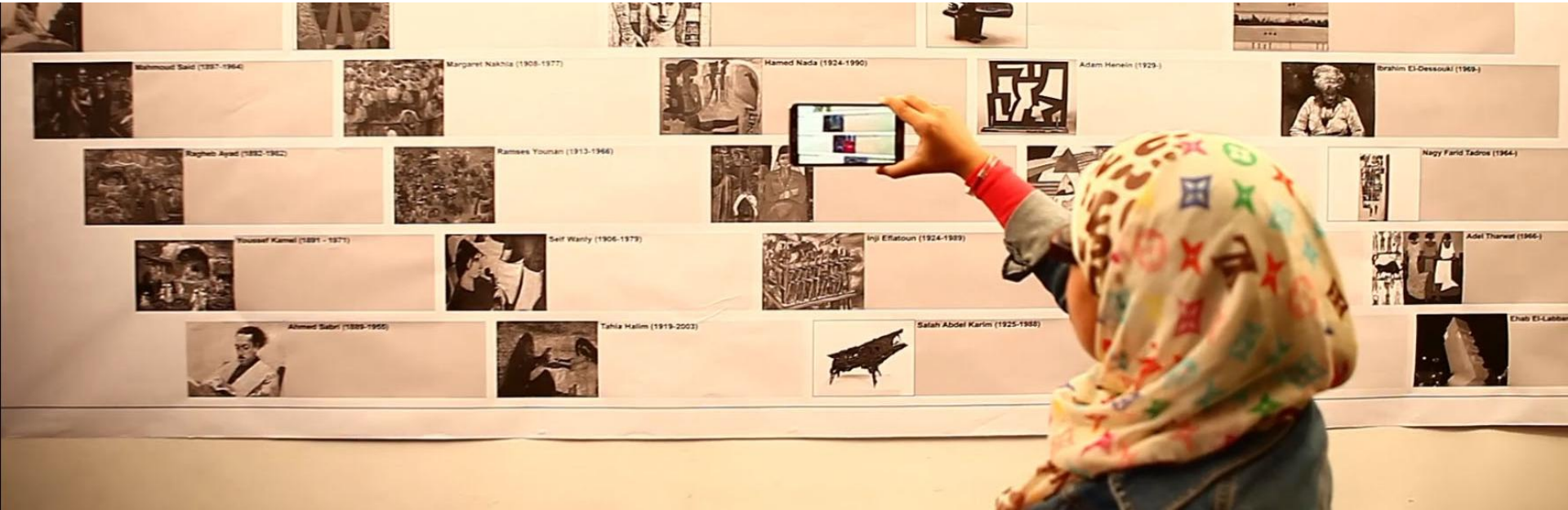
The Wall of Knowledge

Exploring the Egyptian Art History through Augmented Reality

Fathi Saleh, Mohammad Nabil, Nessma Sameer

CULTNAT - Bibliotheca Alexandrina, Egypt

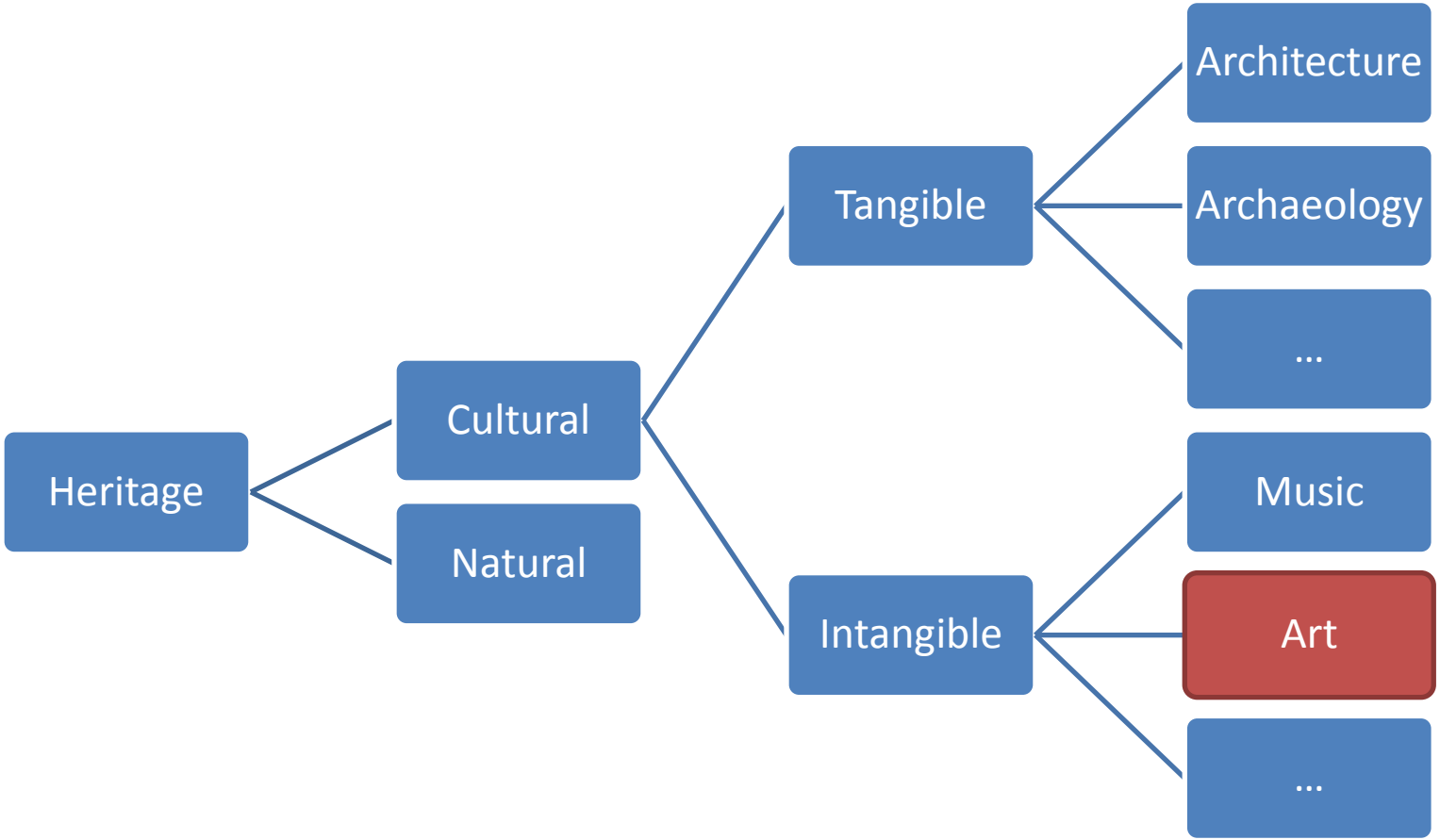


CULTNAT

- The Center for Documentation of Cultural and Natural Heritage
- One of Bibliotheca Alexandrina research centers
- Cairo, Egypt
- Aim:
  - applying the latest technological innovations to document Egypt’s cultural and natural heritage



Classification



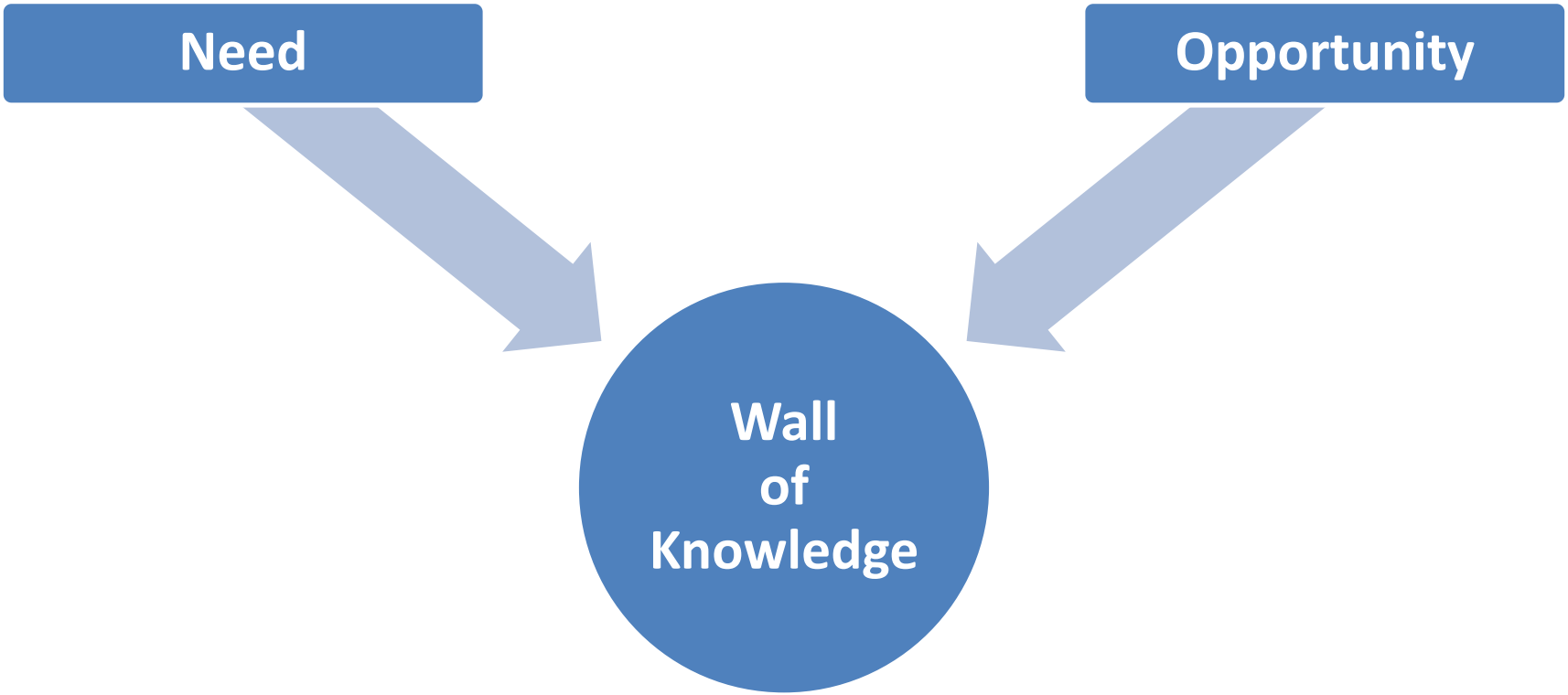
Technologies

Documentation

- Panoramic Photography
- Photogrammetry
- Laser Scanning

Dissemination

- Location Based apps
- Virtual Reality
- Augmented Reality

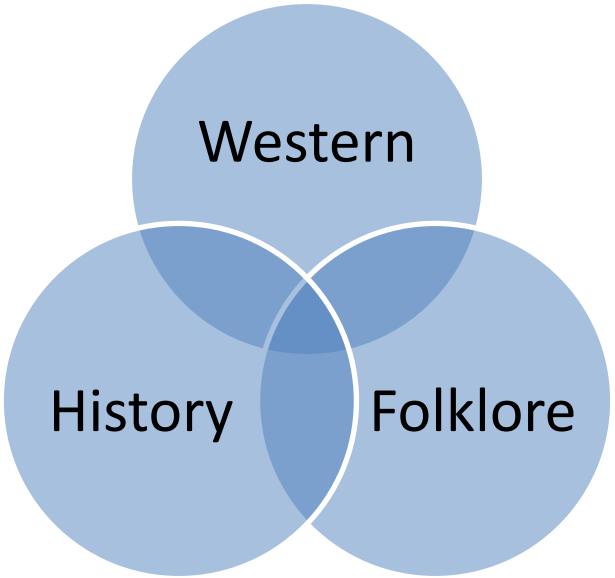

Need
Opportunity
Wall
of
Knowledge

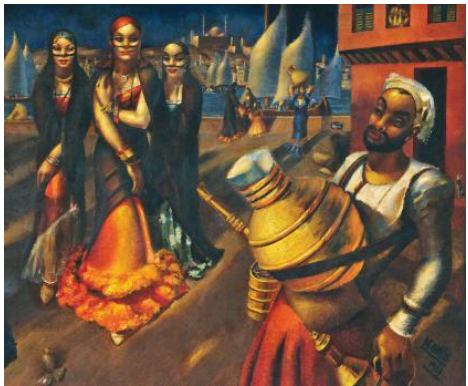
Need

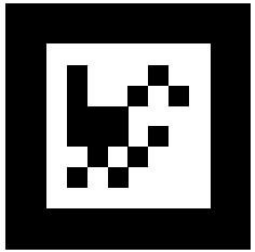
- **Salon d’automne 2014**
- **Egyptian Plastic Art From 1890**





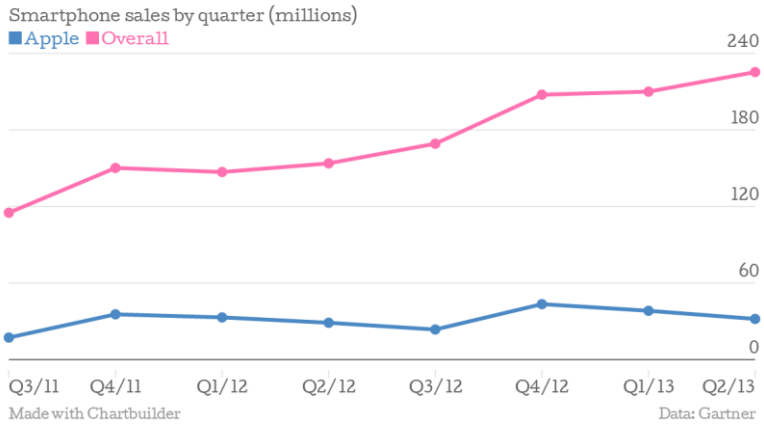








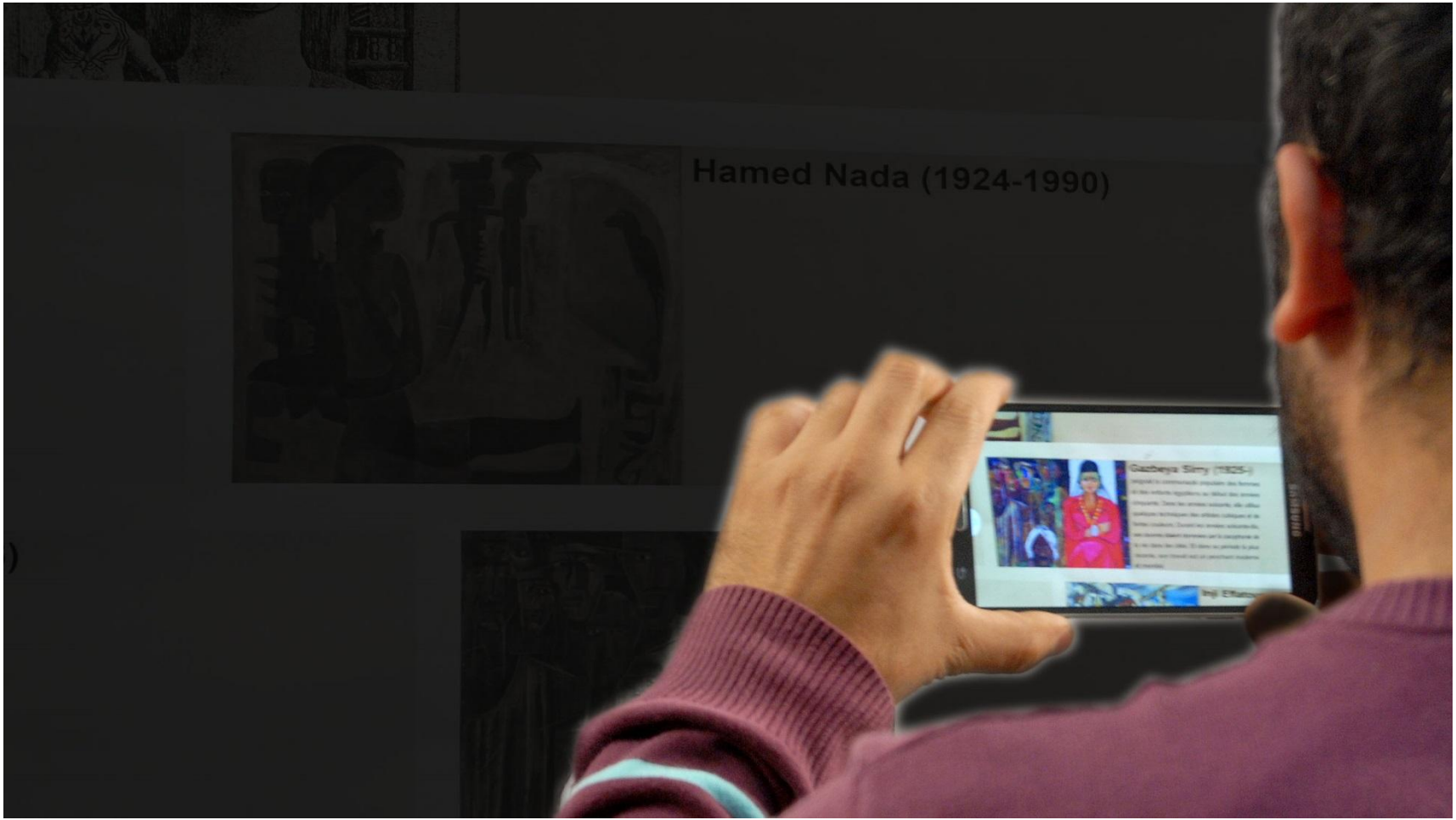
Opportunity






Hamed Nada (1924-1990)


Hamed Nada (1924-1990)


Hamed Nada (1924-1990)

Génération des Pionniers



Mahmoud Mokhtar (1891-1934)



Mahmoud Said (1897-1964)



Ragheb Ayad (1892-1982)



Youssef Kamel (1891 - 1971)



Ahmed Sabri (1889-1955)

Deuxième Génération



Effat Nagy (1905-1994)



Margaret Nakhla (1908-1977)



Ramses Younan (1913-1966)



Seif Wanly (1906-1979)



Tahia Halim (1919-2003)

Troisième Generation



Abdel-Hadi El-Gazzar (1925-1966)



Hamed Nada (1924-1990)



Gazbeya Sirry (1925-)



Inji Eflatoun (1924-1989)



Salah Abdel Karim (1925-1988)

Quatrième Generation



Sobhi Girgis (1929-2013)



Adam Henein (1929-)



Mostafa Abdel Moati (1938-)



Mostafa El-Razzaz (1942-)



Ahmed Nawar (1945-)

Cinquième Generation



Ayman El-Semari (1965-)



Ibrahim El-Dessouki (1969-)



Nagy Farid Tadros (1964-)



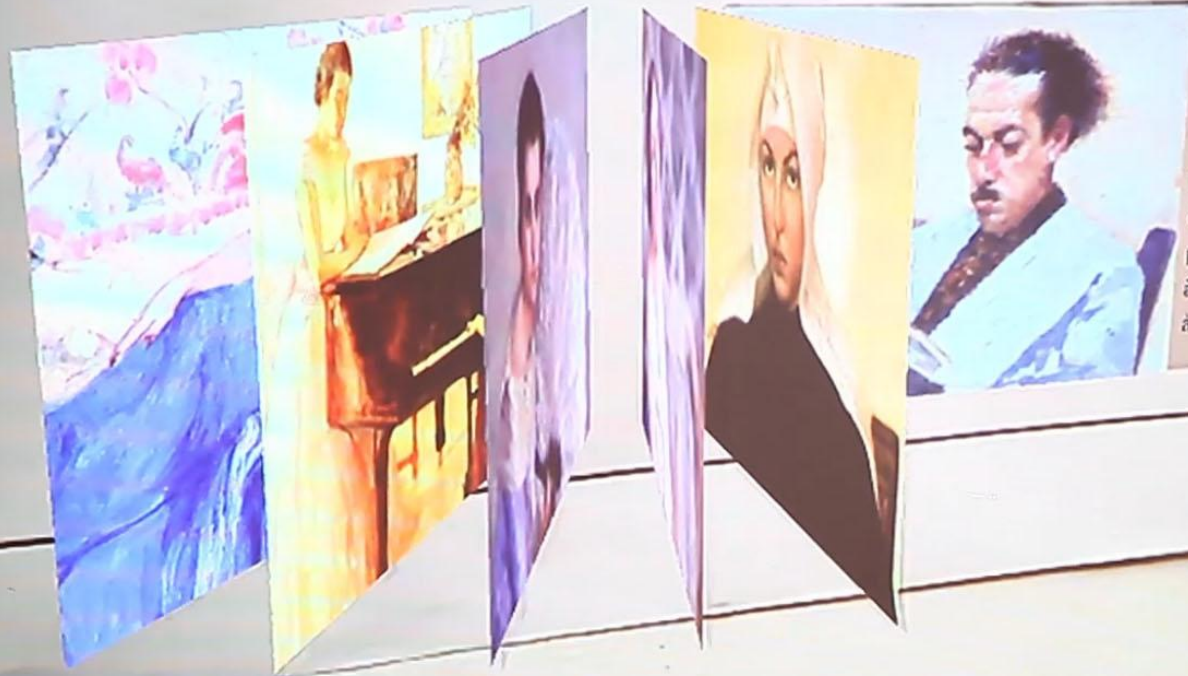
Adel Tharwat (1966-)



Ehab El-Labban (1976-)

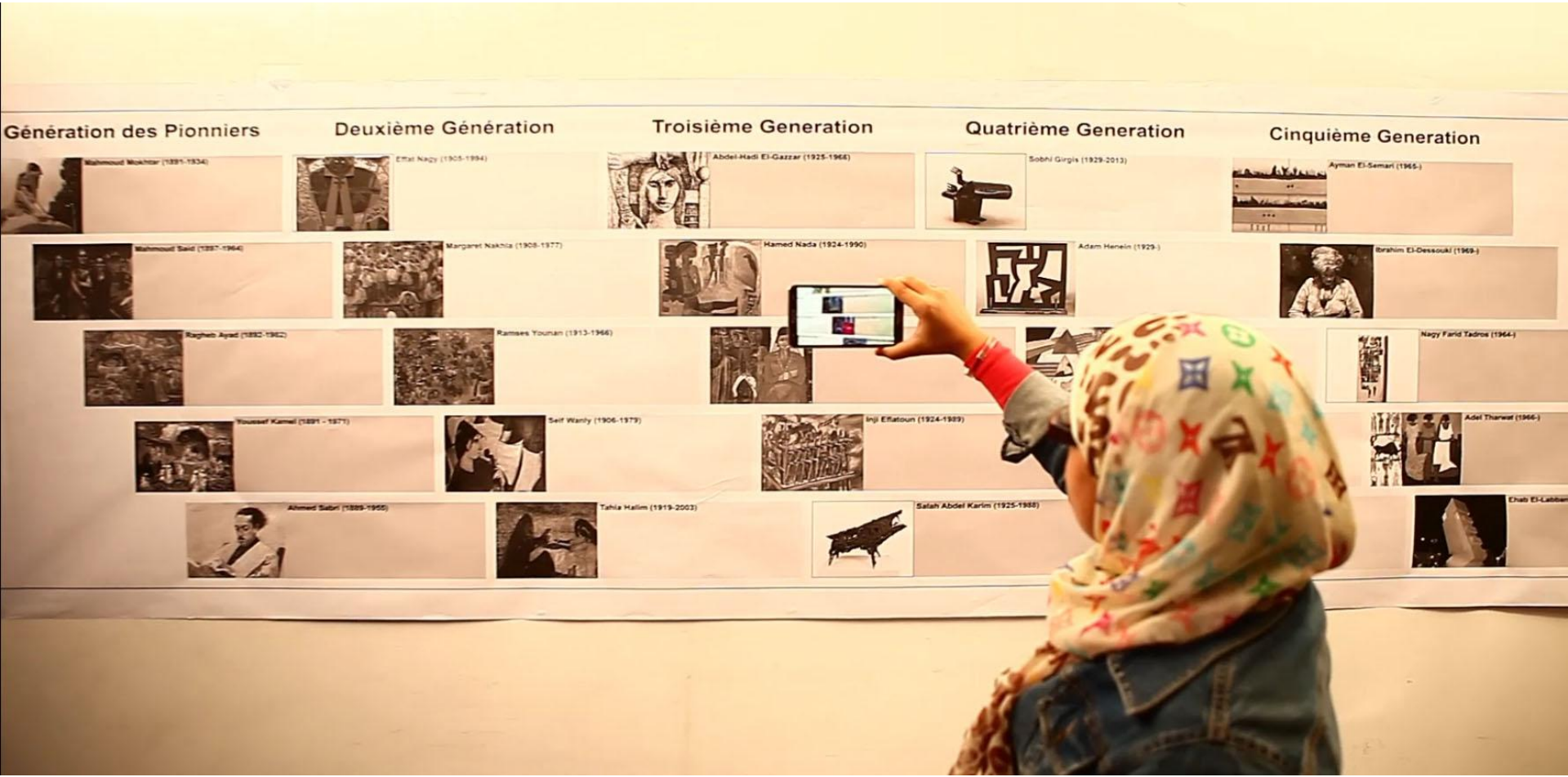
Le Mur de la Connaissance: La Chronologie de L'Art Plastique Pendant le 20e Siecle

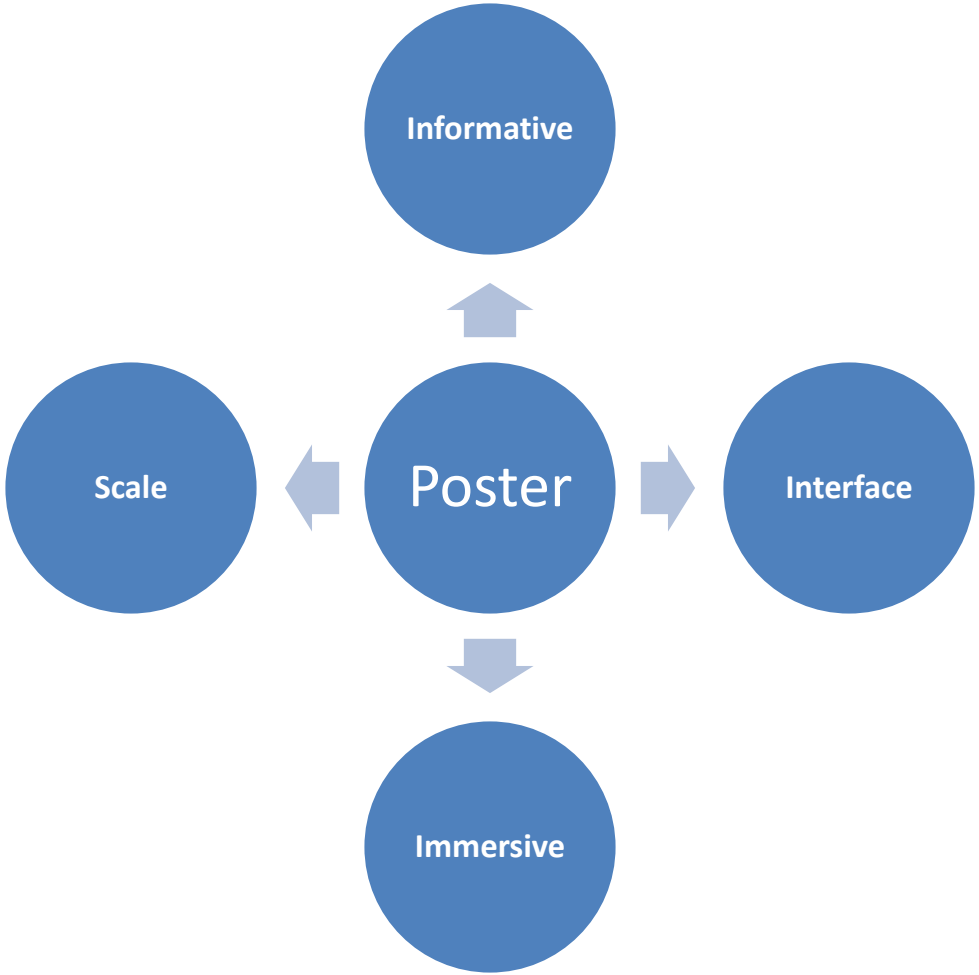




Ahmed Sabri (1889-1955)

est le maitre portraitiste qui utilisa les huiles, les pastels et les aquarelles dans les portraits magistraux de sa femme, des artistes et des figures littéraires tels que l'écrivain Tawfik El Hakim de même il est le professeur de la plupart des portraitistes de la deuxième génération. Il a étudié à l'école des Beaux Arts à Paris après ses études à l'école des Beaux Arts au Caire.


Génération des Pionniers
Deuxième Génération
Troisième Generation
Quatrième Generation
Cinquième Generation
Mahmoud Mokhtar (1891-1934)
Effat Nagy (1905-1994)
Abdel-Hadi El-Gazzar (1925-1966)
Sobhi Girgis (1929-2013)
Ayman El-Semari (1965-)
Mahmoud Said (1897-1964)
Margaret Nakhla (1908-1977)
Hamed Nada (1924-1990)
Adam Henein (1929-)
Ibrahim El-Dessouki (1969-)
Ragheb Ayad (1892-1962)
Ramses Younan (1913-1966)
Nagy Farid Tadros (1964-)
Youssef Kamel (1891 - 1971)
Seif Wanly (1906-1979)
Inji Efflatoun (1924-1989)
Adel Tharwat (1966-)
Ahmed Sabri (1889-1955)
Tahia Halim (1919-2003)
Salah Abdel Karim (1925-1988)
Ehab El-Labban


Informative
Scale
Poster
Interface
Immersive

CULTURAMA




CULTURAMA

Compared to CULTURAMA

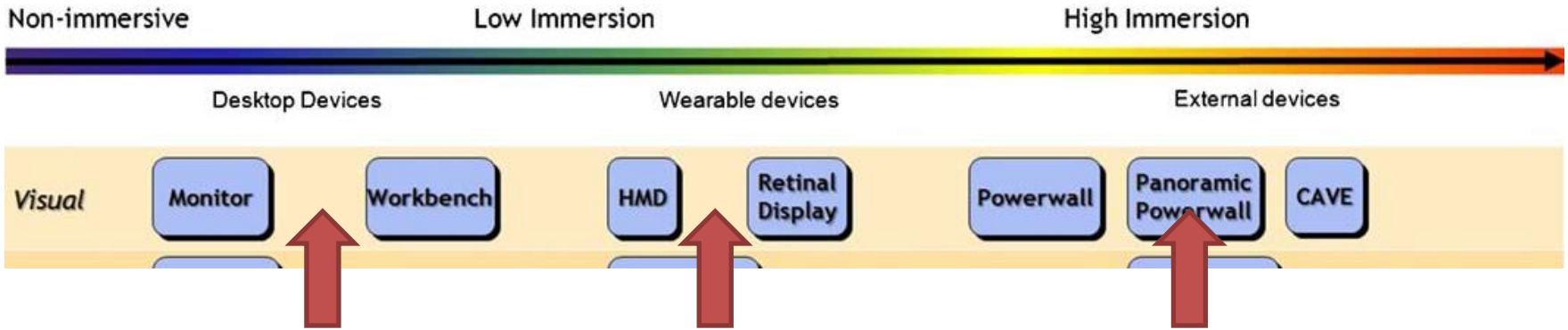
- Immersion
- User Interaction
- Cost
- Content

Level of Immersion

Beyond virtual museums: Experiencing immersive virtual reality in real museums

Marcello Carrozzino, Massimo Bergamasco

Journal of Cultural Heritage, 2010



Wall of Knowledge Mobile

Wall of Knowledge Glass

CULTURAMA

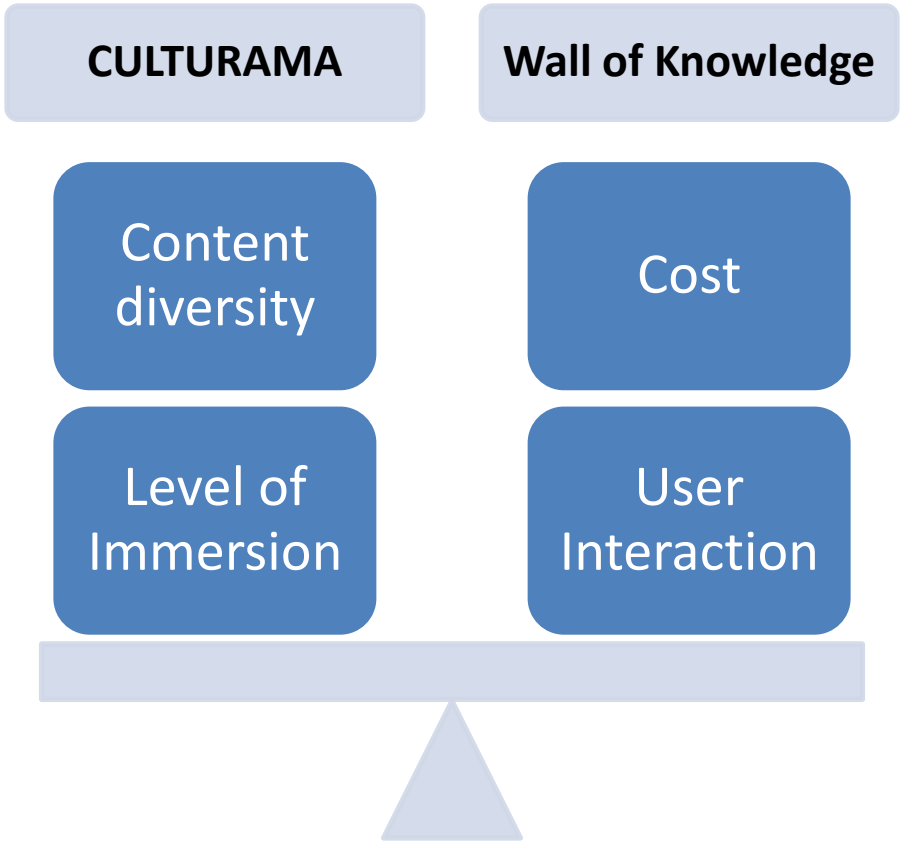
CULTURAMA	Wall of Knowledge
• Controlled by the presenter	• Controlled by users • One user/meter = 10 users/CULTURAMA • Can be personalized
• Good as a lecturing tool	• Better user experience • More informative





Compared to CULTURAMA

CULTURAMA	Wall of Knowledge
• Expensive equipment	• Personal mobile devices
• Diversity of content	• Single topic only


CULTURAMA
Wall of Knowledge
Content diversity
Cost
Level of Immersion
User Interaction

Testing & Questionnaire

- 26 volunteers
- From different backgrounds
- Tablets, mobiles & glasses
- Questionnaire





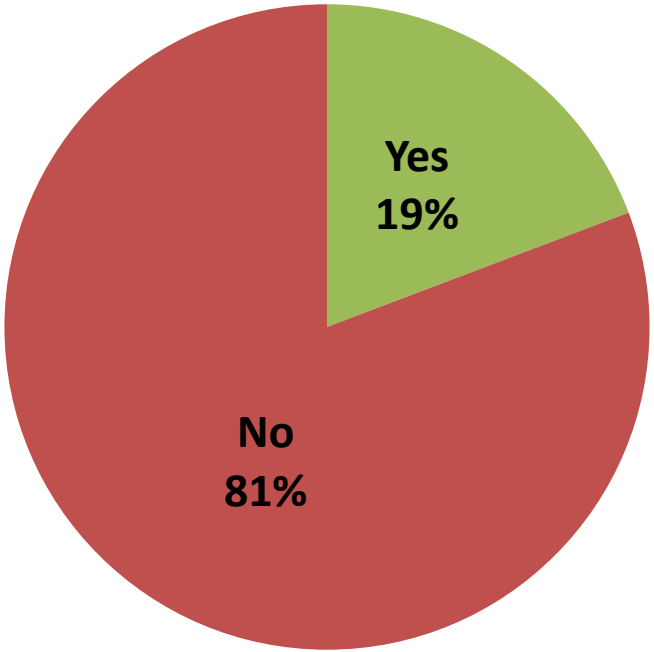






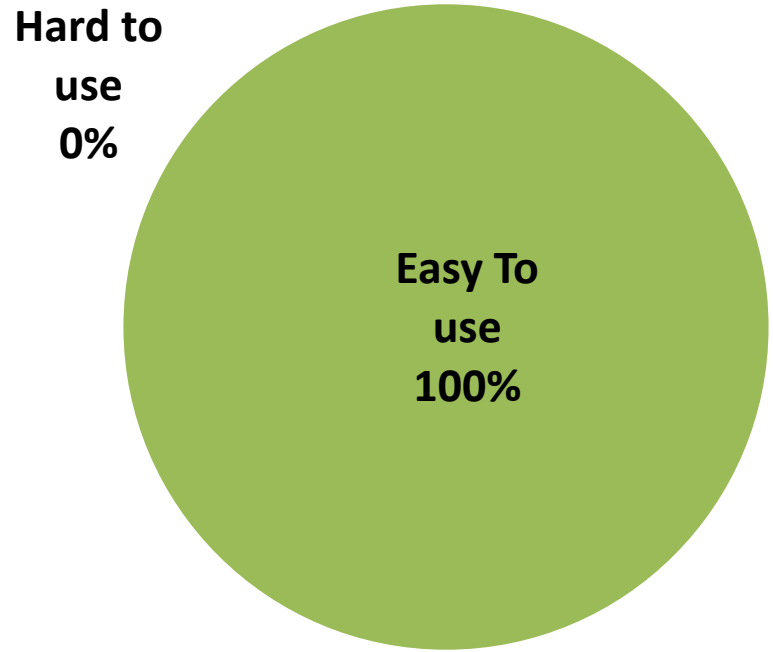
Questionnaire

Have you tried any Augmented Reality applications before?



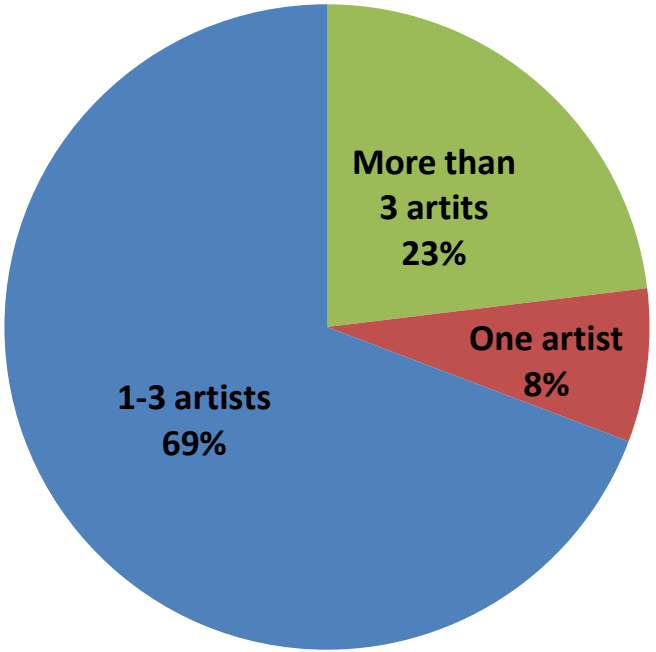
Questionnaire

Was it easy to use the app?



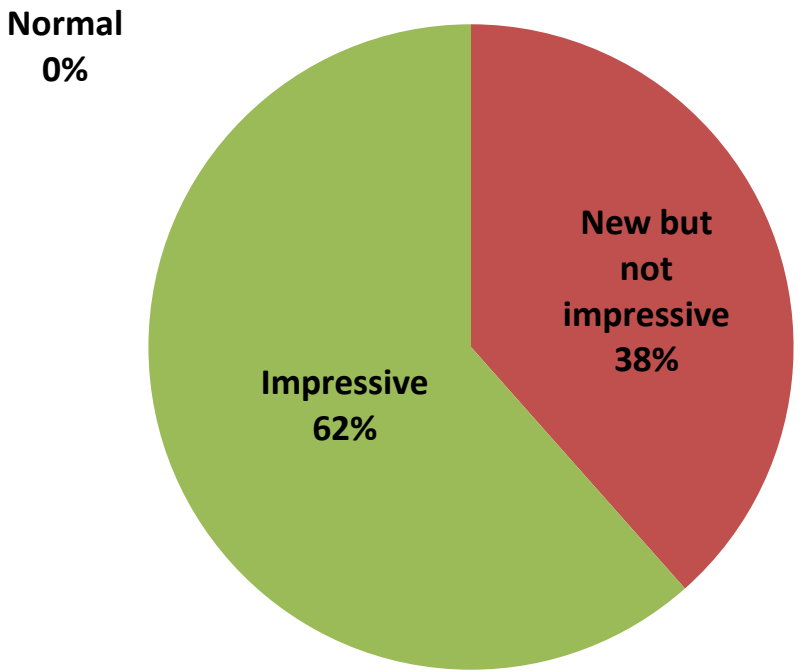
Questionnaire

How many artists did you visit?



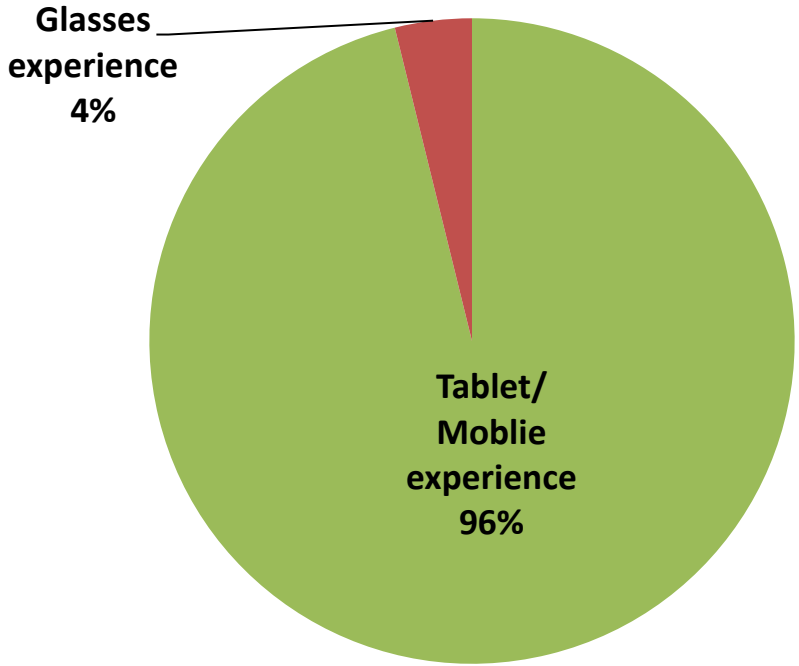
Questionnaire

How do you find the "Wall of Knowledge" idea?



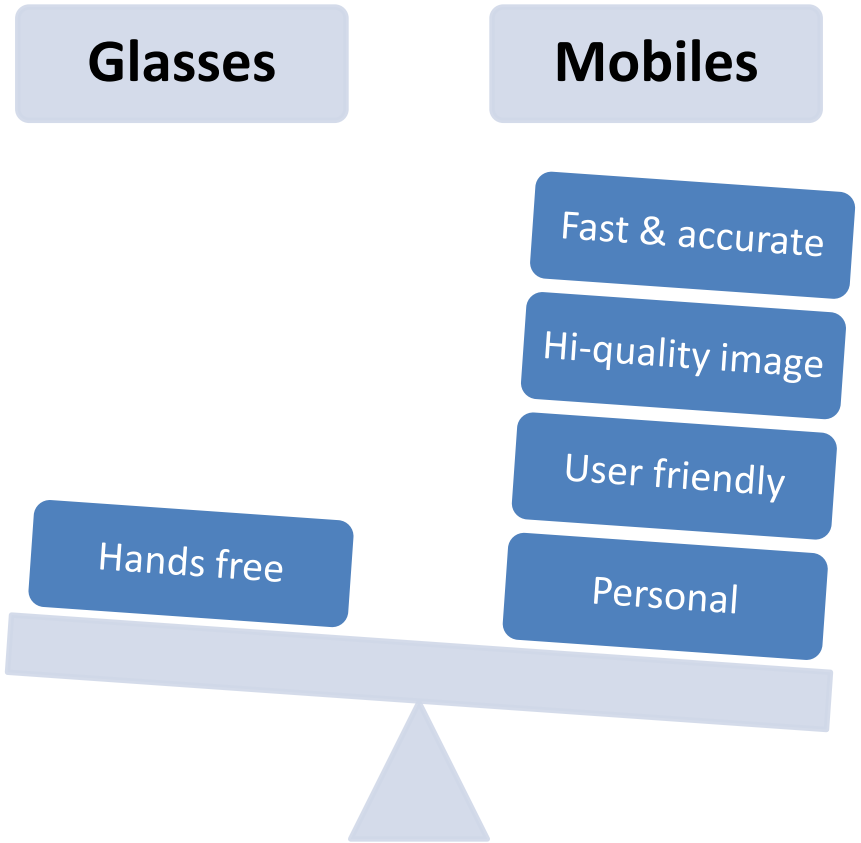
Questionnaire

Which experience is better?



Questionnaire

Why mobiles are better?



Questionnaire

- **Any suggestions?**
  - Startup guidance
  - Voice narration
  - Save experience
  - More content
  - Arabic version
  - Compare paintings
  - Enable it in public squares
  - Educational use

- Zoom in



Feature Plans

- Longer wall
- Variety of augmentation
- Distance-based augmentation
- More glasses
- Salon d'automne 2015
- Bibliotheca Alexandrina

Conclusion

The Wall of Knowledge is a new concept that tries to exploit AR to increase the public engagement with heritage content.

Preliminary indicators say it might be a good method for the dissemination of the Egyptian heritage.



